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Large language models lose diabetes stigma signals when patient culture shifts

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Abstract

A disease label does not always define a stable language model task. We test this problem in diabetes, a common and stigmatized condition where Type 1 and Type 2 communities share a diagnosis but use different cultural language for daily care and public judgment. We built a patient discourse corpus with a manually annotated reference set of 1,526 posts from condition specific Type 1 and Type 2 forums, supported by 1,386 additional texts from patient interviews, forum posts, and published patient narratives for model development. Our lexical and embedding analyses identified distinct diabetes language patterns across the two communities. A stigma detector trained on Type 2 discourse lost sensitivity when applied to Type 1 stigma posts, with F1 falling from 0.91 in domain to 0.43 out of domain. Direct LLM prompting showed that stigma detection varied with prompt context, with generic prompts missing rare stigma signals and calibrated prompts recovering positive class sensitivity. We then introduced a culture sensitive calibration and adjudication framework that does not hard code diabetes type into prediction. Prompt calibration raised GPT-5.4 Macro F1 from 0.515 to 0.776, and importance weighted adjudication improved downstream binary and subtype Macro F1 to 0.767 and 0.368. Our results show that a shared diagnosis does not define a shared language model task, and that culture sensitive calibration with adjudicated expansion can improve the validity of inference from patient language.

Introduction

A disease label does not always define a stable language model task. This assumption can work for biomedical content, but it becomes less reliable for social meaning. Patients can share a diagnosis and still use different words, examples, and interpretive frames to describe daily life with that condition. As a result, aggregate model performance can look strong, even when our model misses meanings specific to a patient community. We use patient culture to mean the community organization of illness language: the vocabulary, treatment references, public stereotypes, and shared assumptions through which patients describe a condition.^{1,2} This is not a claim that every person in a community speaks in the same way. We argue that patient communities can make some meanings more visible than others. We use loss of signal to describe model output patterns, not internal model states. Our analysis focuses on model behaviour under a defined task, not a claim about model intention or belief.^{3,4}

Illness talk is interpreted through the shared norms and frames of the community in which it appears. Words that may seem neutral to outsiders can signal blame, fear, or stigma to those within that community.² This has direct implications for large language models (LLMs), as fluent text does not al-

ways preserve the cultural context that gives patient language its meaning. Recent work shows that these systems reproduce dominant linguistic patterns in ways that affect interpretation in health settings.⁵⁻⁸ One consequence is algorithmic erasure,⁹ where culturally specific meanings weaken during model interpretation and become invisible to automated analysis. In online diabetes discourse, phrases such as #DiabetesOnAPlate can express moral judgment that goes beyond a simple discussion of food.¹⁰ A model without community context may read stigmatizing language as ordinary diabetes talk. For this reason, a disease label is not always a reliable grouping unit for language modeling when the outcome of interest is social meaning as much as physiology. An aggregate benchmark may appear strong and still favor the patient discourse that dominates the pooled sample. The model does not need to produce an explicitly biased answer to lose an equity relevant signal. It can weaken that signal by treating community specific patient language as ordinary text, a risk increasingly recognized in work on medical AI generalization and LLM related inequity.^{11,12}

Diabetes is a strong case for studying this problem, since high prevalence coincides with well-documented stigma and discrimination. The International Diabetes Federation estimates that 589 million adults aged 20 to 79 were living with diabetes in 2024 and projects 853 million by 2050.¹³ The con-

dition also has important social consequences. An international consensus statement reported that about four in five adults with diabetes experience stigma and one in five experience discrimination. A recent review further associated diabetes stigma with lower self-care, poorer psychosocial outcomes, higher HbA1c, and greater risk of acute and chronic complications.^{14,15} Recent work on diabetes communication now argues that language is part of care and that wording can influence self-perception, healthcare engagement, and self-management.^{16,17} This burden becomes more visible when Type 1 and Type 2 diabetes are compared side by side, since they share a diagnostic label but differ in lived experience and social discourse. Type 2 diabetes discourse is often organized around blame, body weight, food, and personal responsibility, whereas Type 1 diabetes discourse is often organized around insulin dependence, visible devices, public misunderstanding, and suspicion tied to treatment.^{18,19} This distinction has direct implications for our analysis, where a pooled diabetes benchmark can score well overall and still give greater visibility to the discourse that dominates the pooled sample. A model that treats diabetes language as a single domain may perform well in aggregate and still miss community-specific social meaning.

These distinctions become visible in digital spaces where patients describe daily routines, seek advice, and respond to judgment from peers, family, clinicians, and the public. Social media can provide information and support for people with diabetes, but it can also intensify stigma and affect what users feel safe to disclose.²⁰ For example, Instagram use has been linked to 35% higher rates of body dissatisfaction among people with Type 1 diabetes, and stigmatizing content such as *#Diabud-diesFail* trivializes insulin use, with most users experiencing bullying in comment sections.²¹ Qualitative work with young people living with Type 1 diabetes suggests that online stigma can change self-presentation. Some users avoid disclosure to avoid being reduced to the condition or targeted by jokes and misinformed comments such as 'this dessert will give me diabetes'.²⁰ Reddit provides a strong empirical setting for our question. Its condition-specific communities organize discussion around lived experience, which makes the cultural language of each patient group more visible. Its pseudonymous structure also reduces some of the social cost tied to discussing stigmatized experiences.²²⁻²⁴ In diabetes, this community structure separates Type 1 and Type 2 communities in a way that makes distinct semantic worlds easier to detect. This setting captures patient language in spaces where meaning is negotiated beyond the fixed structure of survey items or clinical notes. It gives a direct view of the discourse that generalized models must interpret. Because stigma is produced and amplified in digital communication, computational methods have become an important tool for large scale study of patient discourse. Recent reviews of LLM-based health text classification show that most studies still rely on English clinical text, and patient produced discourse and community language are comparatively under-represented.²⁵

Previous natural language processing (NLP) studies have identified stigmatizing language in clinical documentation and other health related text. Recent work has further improved de-

tection through in context learning and transformer models.^{26,27} Machine learning has also detected stigma in public social media text for other health conditions, which shows that large scale stigma analysis is feasible beyond the clinic. In diabetes, recent LLM work has largely addressed misinformation detection, with much less attention to stigma or to shifts in meaning across patient communities.^{21,28} The open question is how patient culture changes the validity of the language model task. This creates an important limitation, as stigma cannot be reduced to a text label.²⁹ It is a social judgment shaped by who speaks, who listens, and the local discourse of a patient community. Recent medical LLM evaluation has also shown that performance can change with the language of the task, even when the underlying medical content is largely the same.^{30,31} We ask if a related failure can occur even within English-language tasks, when the patient culture behind the text changes. We address that problem in diabetes communities by testing if generalized LLMs can identify behavioural concepts such as stigma when those concepts use different language between Type 1 and Type 2 patient groups. Our paper makes three linked contributions. First, it shows that Type 1 and Type 2 diabetes communities occupy overlapping but distinct language spaces. Second, it quantifies the loss of stigma signal that appears when a model trained on one diabetes culture reads the other. Third, it introduces a culture sensitive calibration and adjudication framework that combines prompt calibration with importance weighted adjudication to reduce part of that loss. Importantly, the framework is not a Type 1 versus Type 2 classifier. We use the Type 1 and Type 2 contrast to expose the failure and to audit recovery. The detector itself is trained to identify stigma in patient language. In this paper, culture becomes both a validation target and a design target for medical language modeling, not a background descriptor. The framework draws on Reddit discourse, deidentified interviews collected with partners, and published patient narratives from public sources to move model development away from abstract definitions and toward patient language as it is used in context. Our findings indicate that generalized LLMs can appear reliable within a pooled diabetes domain while missing stigma when social meaning shifts between Type 1 and Type 2 patient cultures. They also show that culture-sensitive calibration with adjudicated data expansion recovered signals that standard prompting leaves behind.

Results

Diabetes is often modeled as a single clinical domain, but patient communities do not describe its social burden in a shared language. We organized our analysis in three steps. First, we measured language separation between Type 1 and Type 2 diabetes communities. Second, we measured how that separation changed prediction patterns when stigma detectors moved across communities. Third, we evaluated culture sensitive prompt calibration and adjudicated data expansion as two ways to recover part of that lost signal. Our analysis moves from language separation to model failure and then to partial recovery. Figure 1 summarizes our workflow.

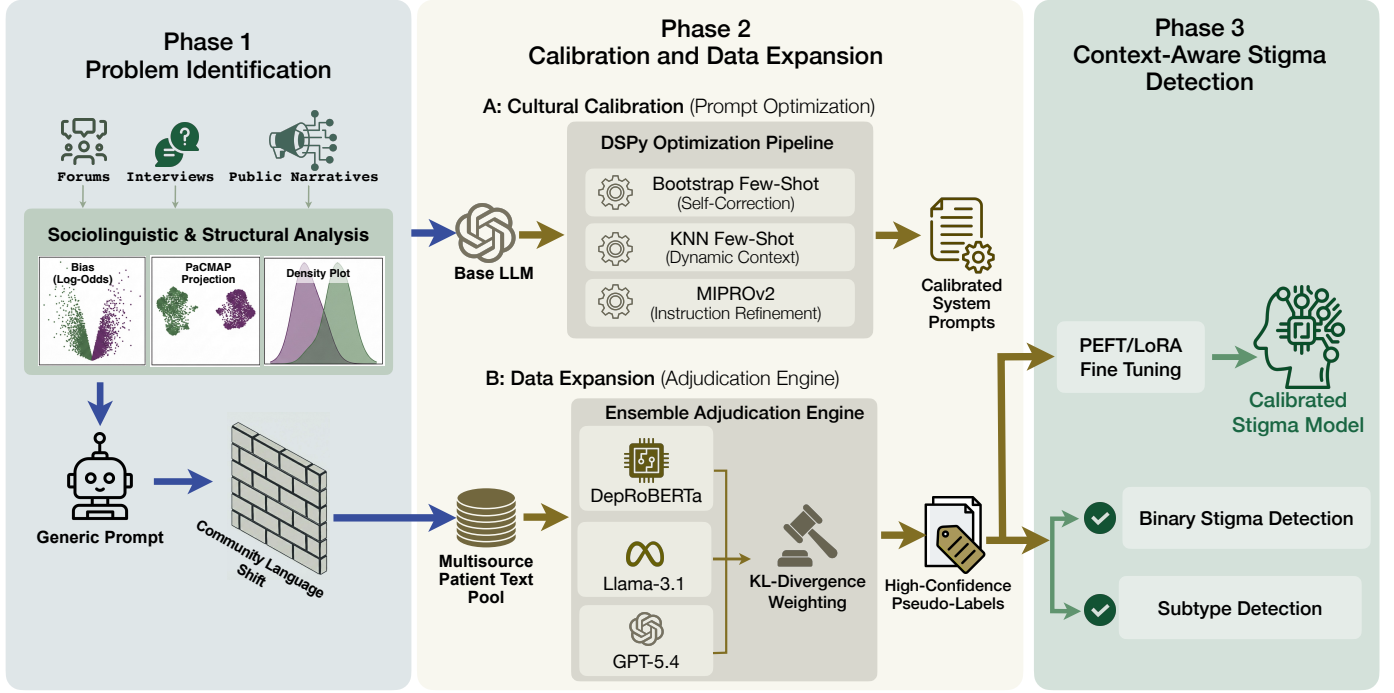


Figure 1: **Study workflow for culture sensitive diabetes stigma detection.** Phase 1 constructs the patient discourse corpus and separates three data roles: the annotated forum reference set, the multisource patient text pool, and the manually reviewed external set. The annotated forum reference set is used to measure language separation and cross community transfer loss. The manually reviewed external set supports an external signal check outside the forum setting. The multisource patient text pool supports data expansion in Phase 2. Phase 2 applies prompt calibration and importance weighted adjudication to improve stigma detection and select retained pseudo labels. Phase 3 fine tunes the downstream detector for binary stigma detection and subtype detection, with diabetes type used only to reveal and audit the cultural shift rather than as the prediction target.

Our manually annotated reference set contained 1,526 posts from condition specific Type 1 and Type 2 diabetes forums. Annotators with backgrounds in health sciences, epidemiology, and health informatics labeled 136 posts as stigma, 1,371 as non stigma, and 19 as unclear. Among the 136 positive posts, internalized stigma was the largest subtype ($n = 88$), followed by experienced stigma ($n = 47$), perceived stigma ($n = 24$), intersectional stigma ($n = 10$), and anticipated stigma ($n = 8$). Subtype labels were not mutually exclusive, and counts can therefore sum above the number of positive posts. Given that stigma posts were rare and the non stigma class dominated the dataset, Macro F1 was the primary model metric, with accuracy reported as a secondary summary. Macro F1 gives similar weight to minority and majority classes, which makes it more informative than accuracy alone for imbalanced text classification.^{32,33} Our subtype distribution clarifies why these model errors have clinical relevance. Prior clinical research has linked internalized and perceived diabetes stigma to distress, social support, and care engagement.³⁴⁻³⁶

Distinct language spaces within diabetes

The two Reddit communities showed distinct patterns in disease-term frequency and in the way their posts were arranged within the embedding space. These patterns were visible in the two-dimensional PaCMAP projection of post embeddings. The 95% kernel density boundaries showed largely separate dense regions for Type 1 and Type 2 posts (Figure 2a). The overlap

between these regions is important, as it shows that the corpora still share a common diabetes vocabulary and do not split into two fully discrete clusters. The Silhouette Score was 0.159 and the Davies-Bouldin Index was 1.858, which supports the same interpretation. These values indicate that the two corpora share enough general diabetes language to overlap and still differ enough to create distinct local vocabularies for model interpretation.

We further observed that the most linguistically significant terms are mutually exclusive to each group (Figure 2b). The data forms a distinct 'V' structure in the volcano plot. This pattern suggests that the two communities are not just using different words, but are talking about diabetes through meaningfully different frames. Type 1 posts centered on insulin dosing and technology, and included terms such as 'bolus', 'pump', and 'omnipod' (Figure 2b). This language is consistent with current clinical guidance that emphasizes glucose monitoring, insulin delivery, device choice, and device education in technology-supported diabetes care.³⁷ Type 2 posts centered on medication, metabolic change, and disease trajectory, with terms such as 'ozempic', 'metformin', 'remission', and 'fasting'. The same divide appears in the social language of stigma. Type 1 stigma is more often tied to visible devices and insulin use, and Type 2 stigma is more often tied to food, weight, and personal responsibility.¹⁷⁻¹⁹ This contrast indicates that Type 1 and Type 2 diabetes should not be assumed to form a single validation domain for stigma detection. The contrast motivates transfer testing and

stratified evaluation, but it does not require the final detector to predict diabetes type.

Signal loss under community transfer

Beyond differences in vocabulary, language separation also defines what becomes visible to the model. Weighted log-odds analysis showed that generic diabetes language did not have a neutral midpoint between the two communities (Figure 3a). Type 1 markers such as 'bolus' (log-odds +4.7) and 'omnipod' (+4.2) were far from Type 2 markers such as 'metformin' (-3.9) and 'remission' (-3.1). More clinically ordinary words also aligned more closely with Type 2 discourse. The terms 'eating' (-0.8), 'sugar' (-0.5), and 'blood' (-0.5) all had 95% confidence intervals that excluded zero. This distinction affects interpretation of model behaviour. A model that reads these terms as general diabetes language may learn a Type 2-centered default and then read Type 1 device and dosing language as a less familiar variant. Here, cultural shift is measured through community-specific language use, including vocabulary, framing, and the relative salience of common diabetes terms. The transfer experiment tests if stigma sensitivity is preserved when those community-specific cues change.

The transfer test showed how this imbalance affects prediction (Figure 3b). For confirmed Type 1 stigma posts, the in-domain model assigned a mean stigma probability of 0.74 and reached recall of 1.00. The out of domain model assigned a mean probability of 0.47 to the same posts. The distributional difference was large (Kolmogorov-Smirnov statistic = 0.82, $p < 0.001$), and the F1 score fell from 0.91 to 0.43. Medical AI research has shown that distribution shift can reduce real world performance when the data used in practice differ from the data used during development.^{11,38} Our result identifies a more specific form of this problem. The disease label, platform, and target were held fixed, but failure appeared when the patient culture behind the text changed. This pattern points to an interpretive error, beyond an incidental drop in performance. The model encountered familiar diabetes language, but the stigma signal became less visible once the community context changed. We used the Type 2 to Type 1 transfer test as our main interpretable transfer probe. This test captured the loss of visibility for confirmed Type 1 stigma posts under a model trained on Type 2 language. Our downsampled sensitivity analyses showed the same direction of effect, which supports our interpretation that sample size imbalance alone did not produce this result.

Prompt context and stigma sensitivity

The lexical transfer probe showed that stigma sensitivity changed when the training community changed. We then tested the effect of prompt context on LLM behaviour. Table 1 compares the performance of GPT-5.4 and Llama-3.1-8B for four prompt settings: a generic codebook prompt, a Type 1 calibrated prompt, a Type 2 calibrated prompt, and a pooled calibrated prompt. The same held out posts were used across settings. This design tested the effect of examples and task language on stigma sensitivity across Type 1 and Type 2 posts. For

GPT-5.4, generic prompting and Type 1 calibrated prompting failed to recover positive class signal on Type 1 posts, whereas Type 2 calibrated and pooled prompts recovered positive class F1. On Type 2 posts, Type 2 calibrated prompting produced the strongest GPT-5.4 Macro F1. Llama-3.1-8B showed a smaller difference between prompt settings, with positive class sensitivity remaining more stable across prompt conditions. Relative to the generic codebook prompt, exact McNemar tests did not show detectable paired changes in overall correctness for the Type 1 calibrated prompt on Type 1 posts ($p = 1.0$), the Type 2 calibrated prompt on Type 1 posts ($p = 0.6875$), the Type 2 calibrated prompt on Type 2 posts ($p = 0.092$), or the Type 1 calibrated prompt on Type 2 posts ($p = 0.5$). These results link LLM stigma detection to prompt context, but they do not show that a Type 1 prompt always performs best on Type 1 posts or that a Type 2 prompt always performs best on Type 2 posts.

Culture calibrated prompting

Next, we evaluated whether a general LLM could detect stigma more reliably when community context was included directly in the task definition. Prompt calibration tested if task-specific instructions and examples could help the model identify stigma signals missed under generic prompting. Table 2 compares GPT-5.4 with Llama-3.1-8B on binary stigma detection.^{39,40} The zero-shot GPT-5.4 result was high in accuracy (0.917) but weak on the positive stigma class ($F1(+) = 0.074$), which shows why accuracy alone would be misleading in this dataset. MIPROv2 raised GPT-5.4 Macro F1 from 0.515 to 0.776 and raised $F1(+)$ from 0.074 to 0.593. As a secondary paired test of overall correctness, exact McNemar's test did not show a detectable change for GPT-5.4 relative to zero shot prompting ($p = 0.701$). This pattern is consistent with a main gain from rare positive class recovery, with little change in aggregate accuracy. Bootstrap Few-Shot and Bootstrap Random improved recall but produced lower Macro F1, whereas Bootstrap Optuna preserved high negative class performance and missed more positive stigma cases. For Llama-3.1-8B, MIPROv2 also produced the highest Macro F1 (0.718) and the highest positive class F1 (0.507). Exact McNemar's test showed a detectable change in overall correctness relative to zero shot prompting ($p = 1.64 \times 10^{-10}$). These results are in line with previous prompt optimization work showing that jointly tuning instructions and demonstrations can improve language model programs without changing model weights.^{41,42}

Adjudicated data expansion

We used the second calibration strategy to explore if unlabeled posts could be incorporated safely into training when model disagreement identified cases with uncertain cultural interpretation. Table 3 reports the performance of Llama-3.2-3B fine tuned on different training sets. The original labeled set produced high accuracy (0.930) but low positive class F1 (0.222), which indicates that the model learned the majority non-stigma class better than the minority stigma class. Majority voting improved binary Macro F1 to 0.724 and multi-label Macro F1 to 0.279. The ensemble-adjudicated set improved

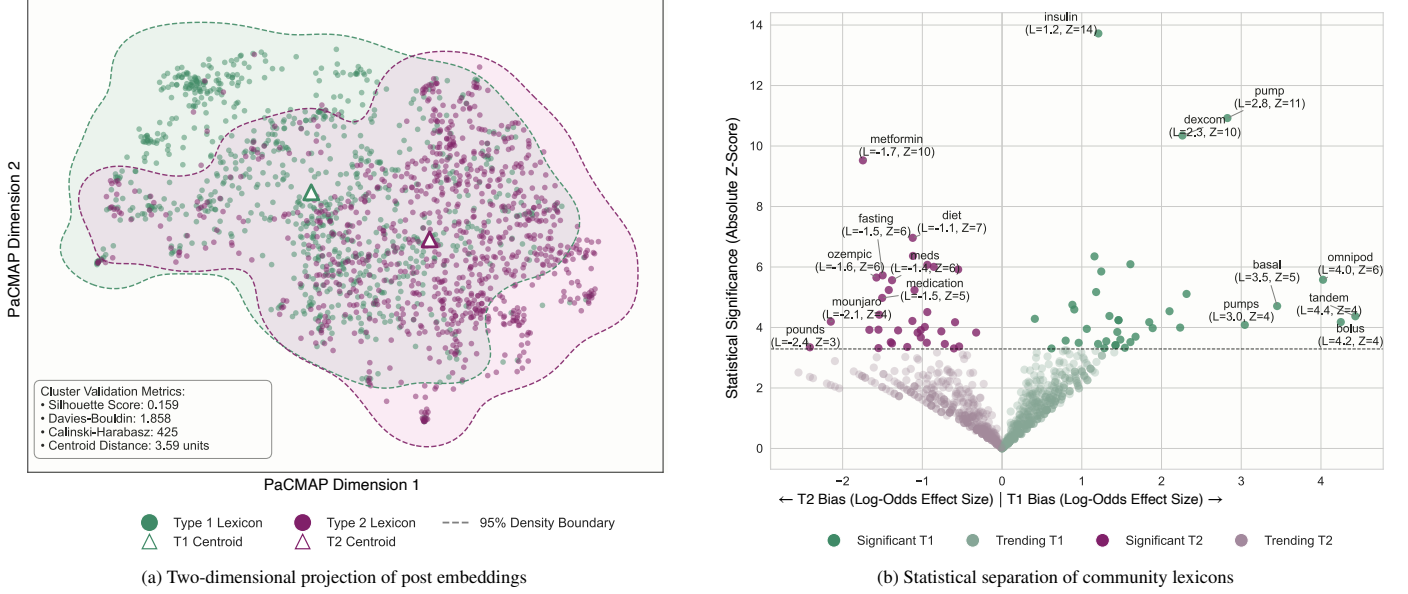


Figure 2: **Language separation between Type 1 and Type 2 diabetes communities.** (a) PaCMAP projection of post embeddings with 95% kernel density boundaries. The communities partially overlap but form distinct dense regions. (b) Weighted log odds analysis identifies terms associated with each community. Type 1 terms center on insulin dosing and devices, whereas Type 2 terms center on medication, metabolism, and disease trajectory.

Table 1: **Direct LLM prompting across diabetes communities.** The table compares generic, community calibrated, cross community, and pooled prompt settings for GPT-5.4 and Llama-3.1-8B. The same held out posts were used across prompt settings. We used diabetes type only for stratified evaluation.

Model	Prompt setting	Type 1 M-F1	Type 1 F1(+)	Type 2 M-F1	Type 2 F1(+)
GPT-5.4	Generic codebook	0.453	0.000	0.479	0.000
GPT-5.4	Type 1 calibrated	0.4485	0.000	0.472	0.000
GPT-5.4	Type 2 calibrated	0.548	0.21	0.608	0.316
GPT-5.4	Pooled calibrated	0.569	0.235	0.593	0.267
Llama-3.1-8B	Generic codebook	0.667	0.467	0.580	0.25
Llama-3.1-8B	Type 1 calibrated	0.625	0.45	0.596	0.3
Llama-3.1-8B	Type 2 calibrated	0.609	0.45	0.516	0.27
Llama-3.1-8B	Pooled calibrated	0.591	0.449	0.583	0.333

binary Macro F1 to 0.767 and multi-label Macro F1 to 0.368, which was the strongest result for both primary tasks. GPT-5.4 labels alone performed well for binary detection (Macro F1 = 0.751) but remained below the adjudicated set on multi-label subtype detection (Macro F1 = 0.319). As secondary paired tests of overall correctness relative to the original labeled training set, exact McNemar tests showed detectable changes for binary detection with majority voting ($p = 4.18 \times 10^{-11}$), ensemble adjudication ($p = 2.12 \times 10^{-13}$), and GPT-5.4 labels alone ($p = 6.96 \times 10^{-13}$). For subtype detection, exact McNemar tests showed a detectable paired change for the ensemble-adjudicated set ($p = 0.049$), but not for majority voting ($p = 0.26$) or GPT-5.4 labels alone ($p = 0.167$). Subtype classification was more challenging than binary detection in this corpus, where stigma posts were sparse, subtype categories overlapped, and brief posts often lacked enough context to distinguish closely related forms of stigma.

These results show the role of adjudication in semi-supervised learning. Pseudo-labeling can expand training data, but it can also amplify early errors when labels are accepted without checks.⁴³ Agreement among models alone did not pro-

vide the most reliable basis for data expansion. Performance was strongest when samples were included only after the adjudication rule jointly considered model decisiveness and uncertainty. In this setting, disagreement functioned as a signal of unstable cultural interpretation, extending beyond label uncertainty. It identified posts where accepting pseudo-labels without adjudication would be more likely to weaken the training signal. The gain from data expansion was tied to the quality of the added labels, where the model improved when the expanded set added linguistically varied examples without diluting the stigma signal.

The recovery framework did not use diabetes type as a prediction target. We therefore audited gain preservation across communities by stratifying held out predictions after inference. This audit tested the distribution of the improvement across Type 1 and Type 2 posts. The ensemble-adjudicated model improved Macro F1 in both communities and reduced the Type 1 to Type 2 performance gap from 0.24 to 0.165. Since this gap is derived from stratified F1 values, we did not use McNemar's test to test the gap itself. As a secondary paired test of overall correctness on the same stratified held out subset, exact Mc-

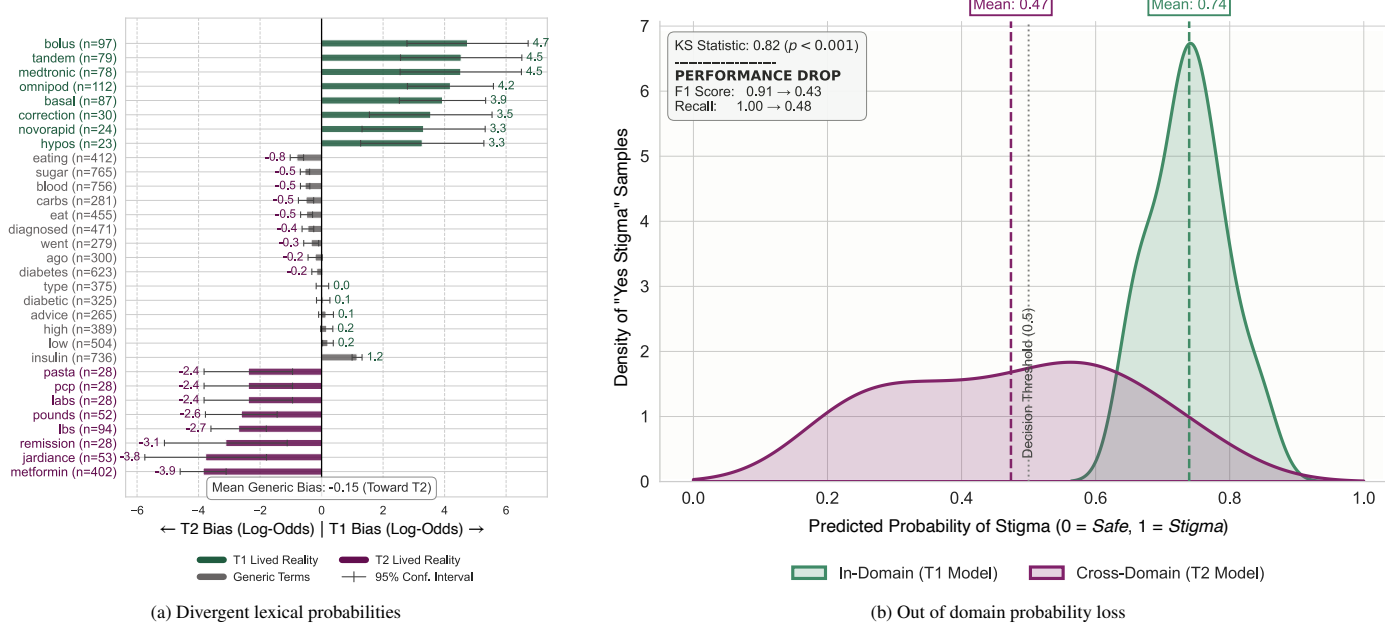


Figure 3: **Loss of stigma signal under community transfer.** (a) Weighted log-odds ratios for terms associated with Type 1 and Type 2 diabetes discourse. Gray bars indicate common diabetes terms that align more closely with Type 2 discourse in this corpus. (b) Predicted stigma probabilities for confirmed Type 1 stigma posts under the in-domain and out of domain classifiers. The out of domain model shifts many confirmed stigma posts toward the decision boundary (Kolmogorov-Smirnov statistic = 0.82, $p < 0.001$).

Table 2: **Prompt optimization results for binary stigma detection.** The table reports binary stigma detection performance for GPT-5.4 and Llama-3.1-8B. M-Prec, M-Rec, and M-F1 denote macro precision, macro recall, and macro F1. F1(+) denotes the stigma class, and F1(-) denotes the non-stigma class.

Method	GPT-5.4						Llama-3.1-8B					
	Acc	M-Prec	M-Rec	M-F1	F1(+)	F1(-)	Acc	M-Prec	M-Rec	M-F1	F1(+)	F1(-)
Baseline (Zero-shot)	0.917	0.958	0.519	0.515	0.074	0.957	0.755	0.594	0.744	0.594	0.339	0.850
MIPROv2	0.927	0.767	0.786	0.776	0.593	0.960	0.878	0.680	0.811	0.718	0.507	0.930
Bootstrap Few-Shot	0.874	0.676	0.809	0.714	0.500	0.928	0.868	0.673	0.823	0.712	0.500	0.924
Bootstrap Random	0.874	0.676	0.809	0.714	0.500	0.928	0.831	0.636	0.786	0.664	0.427	0.901
KNN Few-Shot	0.884	0.667	0.728	0.690	0.444	0.935	0.782	0.617	0.793	0.628	0.389	0.867
Bootstrap Optuna	0.927	0.963	0.577	0.614	0.267	0.962	0.626	0.572	0.726	0.514	0.280	0.747

Nemar's test showed a detectable change relative to the tuned original set ($p = 0.0059$). The stratified audit indicates that the framework responds to cultural shift without hard coding Type 1 and Type 2 as separate model branches.

External patient narrative check

Our final performance estimates came from the manually annotated forum reference set. To test whether the calibrated model also retained signal outside the forum setting, we conducted a patient narrative check using 150 published patient narratives. Two annotators reviewed these texts with the same stigma codebook. Our goal was not to estimate population prevalence, but to explore if stigma expressions in non-forum patient narratives were captured by the same culture sensitive calibration approach. The calibrated model identified stigma in patient narratives with Macro F1 of 0.715 and positive class F1 of 0.476. Compared with the generic codebook prompt on the same 150 narratives, exact McNemar's test showed a detectable paired change in overall correctness ($p = 0.0059$). Errors were

concentrated in short excerpts where the speaker described diabetes management without enough context to distinguish self blame from ordinary self management. This narrative check supports our main result by showing that the calibrated framework retained stigma signal in an external patient discourse setting.

Discussion

A disease label is not always a reliable unit for language modeling. We show that Type 1 and Type 2 diabetes communities can share a diagnosis but use different cultural languages to describe the social burden of that diagnosis. This language difference was visible in the lexicon, in the embedding structure, and in model behaviour. The out of domain transfer test provided the clearest evidence. A stigma detector calibrated to Type 2 discourse assigned much lower stigma probabilities to confirmed Type 1 stigma posts than the in domain model, and F1 fell from 0.91 to 0.43. Direct LLM prompting showed that

Table 3: **Semi-supervised adjudication results.** The comparison of Llama-3.2-3B fine tuned on different dataset configurations. Added samples refer to pseudo labeled posts added to the original manually labeled set.

Training Configuration	Added Samples	Stigma Detection (Binary)						Stigma Type (Multi-label)	
		Acc	M-Prec	M-Rec	M-F1	F1(+)	F1(-)	Acc	M-F1
Baseline (Zero-shot)	—	0.743	0.576	0.708	0.568	0.294	0.843	0.700	0.092
Tuned (Original Set)	0	0.930	0.965	0.563	0.593	0.222	0.963	0.920	0.111
Tuned (Majority Voting)	713	0.923	0.738	0.711	0.724	0.489	0.959	0.913	0.279
Tuned (Ensemble Adjudicated)	712	0.943	0.844	0.722	0.767	0.564	0.970	0.927	0.368
Tuned (GPT-5.4 Labels Only)	756	0.937	0.801	0.718	0.751	0.537	0.966	0.920	0.319

Table 4: **Type stratified evaluation of the recovery framework.** Diabetes type was used only for stratified evaluation, not as a prediction target. Gap represents the absolute difference between Type 1 and Type 2 Macro F1.

Model setting	Overall M-F1	Type 1 M-F1	Type 1 F1(+)	Type 2 M-F1	Type 2 F1(+)	Gap
Zero shot GPT-5.4	0.466	0.453	0.000	0.479	0.000	0.026
MIPROv2 GPT-5.4	0.667	0.654	0.4	0.679	0.4	0.025
Baseline Llama3.2-3B	0.657	0.68	0.5	0.614	0.333	0.066
Tuned original set Llama3.2-3B	0.663	0.585	0.267	0.825	0.667	0.24
Tuned ensemble adjudicated Llama3.2-3B	0.715	0.660	0.4	0.825	0.667	0.165

prompt context changed positive class sensitivity, although it did not reproduce a simple community matched transfer pattern. This failure was not a generic accuracy drop on a difficult task. The model failed when language learned from one patient culture was used to read another.

The transfer result connects diabetes stigma research to a measurable model failure. The stigma of diabetes goes beyond hostile words. It is a judgment embedded in terms about food, body size, devices, insulin, self control, and responsibility. Recent work on diabetes language argues that everyday wording can affect dignity, self management, and care relationships. Moreover, qualitative work shows that people living with Type 1 and Type 2 diabetes describe different patterns of judgment in diabetes talk.^{16,17} Our results show that this language problem becomes a model problem once LLMs are used to classify patient discourse. A pooled diabetes benchmark can appear valid even as it gives greater weight to the discourse that dominates the pooled sample.

Medical AI often frames generalization as a shift between hospitals, scanners, sites, languages, or demographic groups. Recent work has shown that distribution shift can change performance after deployment and that generalization is a major challenge for AI systems used in patient care.^{11,38} Our analysis extends this concern to community language within a single disease domain. The target variable, the broad condition label, and the forum platform were unchanged, but failure came from the cultural organization of meaning. This form of shift is harder to detect than a change in image quality or data source since the input still looks clinically familiar. Terms such as 'blood' and 'sugar' are not out-of-vocabulary. They are familiar words that take on different roles in each community.

The Type 1 and Type 2 comparison makes culture more specific for medical NLP. Culture is often operationalized as nationality, language, or race. These categories are useful starting points, but they do not capture all of culture in patient discourse fully.² The Type 1 and Type 2 split shows that one diagnostic

label can contain more than one interpretive community. This point aligns with recent multilingual medical LLM work, where model performance varies by input language and prompt language.³⁰ Our study shows a parallel problem inside English patient discourse. The boundary is not only between languages. It can also appear between patient communities that share a clinical label but attach different social meanings to treatment and daily management.

The same failure mode extends recent LLM equity work to patient discourse. Current frameworks rightly focus on demographic variables, biased medical answers, adversarial prompts, and human review of equity related harms.^{7,12,44} Our results identify a complementary failure mode. The model does not need to produce an explicitly biased answer to lose an equity relevant signal. The signal can weaken when the model reads local language as ordinary or irrelevant text. For digital medicine, validation should not stop at disease labels or broad patient categories. It should test whether the model can interpret the discourse of the community that produced the text. This does not mean that every model should be split into separate Type 1 and Type 2 detectors. That would solve one version of the problem but create another, since patient culture is not a fixed demographic label and not every post can be reduced to diabetes type. Our framework uses Type 1 and Type 2 differently. The contrast is used to reveal the failure and to audit recovery. The detector itself is trained to identify stigma in patient language. This distinction is important, as the goal is not to classify patient identity, but to preserve the social meaning of patient language when that meaning changes across communities.

The GPT-5.4 results expose a practical limitation for current LLM evaluation. The zero-shot model had high accuracy, but the positive class F1 was only 0.074. This gap shows that accuracy alone can give an inflated view of performance in health tasks with rare social signals. The model mostly assigned posts to the non-stigma class, which was often correct in an imbal-

anced dataset, but it missed the clinically and culturally relevant signal. MIPROv2 improved GPT-5.4 Macro F1 to 0.776 and positive class F1 to 0.593, consistent with prior work on prompt optimization as a way to tune language model programs without changing model weights.^{41,42} For GPT-5.4, exact McNemar's test did not show a detectable paired change in overall correctness relative to zero-shot prompting. This does not contradict the Macro F1 gain, since the main improvement came from recovery of the rare positive stigma class rather than from a large change in aggregate correctness.

The main implication is not that better prompts remove cultural language differences. Prompt calibration improved detection when it gave the model a more specific task definition and linked that definition to community language. That improvement also defines the limit of prompting. If a prompt needs community context to recover the label, community context belongs in the evaluation design. Recent LLM equity guidance makes a similar point for health care deployment, where real world performance should be evaluated with attention to linguistic and cultural bias, access, and oversight.⁴⁴ Our results apply that principle to patient discourse through a concrete test. A model should not be described as valid for diabetes stigma until it has been tested in the Type 1 and Type 2 communities where stigma takes different linguistic forms.

Our study does not only ask whether LLM labels can replace manual labels. It shows that model disagreement can help identify where interpretation is uncertain. Simple majority voting assumes that each model has comparable competence for each post. That assumption becomes weaker when a post relies on community-specific cues. A model can be confident for reasons that do not match the local meaning, and another model can hesitate when the post contains a local meaning outside its learned default.

The ensemble-adjudicated set improved binary Macro F1 to 0.767 and subtype Macro F1 to 0.368. Single-model GPT-5.4 labels performed well for binary detection but produced lower subtype performance. This contrast clarifies the value of adjudication in semi-supervised learning, where pseudo-labeling can amplify confirmation bias when early model errors become training data.⁴³ Our proposed importance-weighted adjudication algorithm reduces this risk in two ways. It uses more than one model family, and it weights labels by decision confidence when the confidence gap is informative. Our adjudication rule is not a substitute for expert annotation. It provides a structured way to select unlabeled examples that can expand a small patient-language dataset without accepting every automated label at face value.

The issue is especially relevant for stigma research since the labels are not simple surface features. Internalized, perceived, experienced, anticipated, and intersectional stigma can overlap in short posts. A post can describe a joke from another person and also show self-directed shame. A model trained only to detect the presence of stigma may learn enough to identify the positive class but miss the social pathway represented by the subtype. The adjudicated set improved subtype detection by adding examples without giving the same level of trust to every automated label.

The clinical contribution of this work is not to diagnose individuals from forum posts or patient narratives. It is to measure social language at a scale that can inform communication, patient education, and community review. Internalized stigma was the most frequent subtype in our corpus. This finding has clinical relevance, as prior work links internalized and perceived diabetes stigma to distress, social support, and care engagement.^{14,15,34-36} A binary label can identify whether stigma is present. Subtype labels show whether the burden is tied to self blame, public judgment, treatment visibility, anticipated rejection, or identity specific disadvantage.

This subtype detail gives our results a clearer practical use. In Type 1 diabetes, posts about devices, alarms, insulin dosing, or public misunderstanding may indicate different communication needs than Type 2 posts about food, body weight, medication, or blame. Prior diabetes stigma research has described similar clinical differences. Type 1 examples include stigma tied to visible devices and insulin use, and Type 2 examples include shame tied to lifestyle judgment.^{10,18,19} A single stigma detector risks collapsing these patterns into one broad output. A culture sensitive detector allows researchers to identify which forms of stigma are present in a given community and how those forms change after public campaigns, changes in diabetes technology, or changes in medication discourse. This distinction is also relevant to policy and community practice. Diabetes Canada connects stigma reduction with efforts to change diabetes-related discourse and promote language that reduces stereotyping and blame.^{45,46}

Our primary contribution is methodological, and it starts from a specific model failure. We do not estimate diabetes stigma prevalence in the general population. Instead, we use Type 1 and Type 2 diabetes to show that a shared diagnosis label can require different language model tasks. This distinction is relevant for tasks where the target is social meaning in addition to biomedical content.

Our study has several limitations. Our manually annotated reference set came from condition specific online forums, and those users are not representative of all people living with diabetes. The corpus is English language and platform specific, and the sampling strategy prioritized high engagement posts rather than a random sample of diabetes discussions. We analyzed public forum text, deidentified interview excerpts, and published patient narratives, not private clinical encounters. Extending these results to clinical notes or clinical decision support will require additional validation. These limitations do not weaken our main comparison. The forum reference set gave us a controlled setting for comparing two patient communities within the same disease label. That design strengthens the internal comparison between Type 1 and Type 2 discourse but limits population generalizability. We therefore do not interpret the corpus as a prevalence estimate for diabetes stigma. We interpret it as a test of whether patient culture can change the language model task when the diagnosis, platform, and target concept remain fixed.

Our analysis also had limitations in label availability and class balance. Stigma posts were rare, and subtype labels were sparse. That class structure limited multi label performance

even after adjudicated expansion. Short posts can also lack enough context to separate perceived stigma from experienced stigma or internalized stigma from anticipated stigma. GPT-5.4 is a proprietary model, and model behaviour can change with version updates, safety changes, or API settings.³⁹ Future studies should report model version, prompt, date of access, and decoding settings, then repeat transfer tests when model versions change.

The same design can be tested in other chronic conditions where disease labels group together patients with different daily practices, technologies, or public stereotypes. It should also test non English diabetes communities and patient groups whose language is influenced by region, insurance status, race, ethnicity, gender, disability, or age. The most direct next step would combine patient partnered annotation with prospective validation in new online communities and clinical communication datasets. That design would test whether culture sensitive calibration improves benchmark performance and the quality of information available to clinicians, researchers, and public health teams.

These findings support a practical requirement for medical language modeling. Patient language should not be treated as a neutral layer added to a disease label. When culture changes, the meaning of the task can change even when the diagnosis stays constant. The practical response is to test transfer between patient communities, calibrate prompts with community context, audit performance across communities, and expand training data through adjudication that respects uncertainty. In diabetes, this issue is directly connected to care, since stigma can influence how people describe symptoms, treatments, self management, and clinical interactions.

Methods

Study design and data sources

We designed this study to test if a single disease label can cover more than one patient language. We focused on Type 1 and Type 2 diabetes since the two conditions share the clinical label of diabetes, but differ in treatment routines, public stereotypes, and the forms of stigma that patients describe in daily life. The manually annotated reference set came from condition specific online forums. A separate unlabeled pool supported model development and adjudicated expansion and included forum posts, deidentified interview excerpts, and published patient narratives. We kept these data roles separate. The reference set was used for final evaluation, and the additional patient texts broadened the range of stigma language available during model development. Diabetes type was used to define the comparison, transfer tests, and stratified evaluation. It was not used as a prediction target in the final binary or subtype models. The target labels were stigma presence and stigma subtype. This distinction allowed us to study if Type 1 and Type 2 discourse changed model behaviour without making diabetes type itself the object of prediction.

The primary corpus contained 1,538 posts, with 538 posts from the Type 1 community and 1,000 posts from the Type 2

Table 5: **Definitions used to classify diabetes-related stigma.** Definitions were adapted from prior diabetes stigma work.¹⁴

Stigma type	Definition used for annotation
Experienced stigma	Direct encounters with diabetes-related stereotyping, blame, exclusion, or unfair treatment.
Perceived stigma	Awareness or belief that negative social views about diabetes exist.
Anticipated stigma	Fear or expectation of being judged or treated unfairly because of diabetes.
Internalized stigma	Adoption of negative beliefs about diabetes, with self-blame or shame.
Intersectional stigma	Diabetes-related stigma that interacts with another stigmatized identity, condition, or social position.

community. We sampled posts from the *hot* feed in each subreddit to include posts receiving visible community attention at the time of collection. After eligibility screening, 1,526 posts were included in manual annotation. Hyperlinks and emojis were removed before model development, while title and body text were analyzed together. We kept words tied to devices, medications, foods, insulin dosing, and self-management because those terms were part of the language differences under study. Moreover, a separate pool of 1,386 unlabeled texts supported semi-supervised expansion. This set included 243 Reddit posts retrieved with the phrase *diabetes stigma*, 653 deidentified patient interview excerpts, and 490 excerpts from public health blogs and research articles. These texts did not contribute to final performance estimates. They supplied candidate pseudo-labels and broadened the range of stigma language available for fine tuning. Results were reported only in aggregate. Usernames, direct identifiers, and URLs were excluded from analytic outputs. Non-Reddit interview excerpts included in the unlabeled pool were deidentified before analysis and used only for model development. Figure 1 summarizes the overall workflow of this study and the connection between corpus construction, transfer testing, prompt calibration, adjudicated data expansion, and downstream fine tuning.

Annotation protocol and reference labels

Stigma is a social judgment, not a keyword pattern that can be assumed to transfer across communities. Supervised modeling needed reference labels based on stigma theory and human review. Manual annotation created the reference labels used for prompt calibration, fine tuning, and final evaluation. Annotators with training in social science and health research used a codebook derived from diabetes stigma literature.¹⁴ The codebook defined five stigma subtypes, shown in Table 5. A positive post could receive more than one subtype label, as a single post can describe multiple stigma processes.

Two annotators independently labeled each post as *stigma*, *no stigma*, or *unclear*. For posts labeled *stigma*, annotators also assigned any relevant subtype labels. A pilot set of 400 posts was used to measure agreement before full annotation. We reported Cohen's kappa and Gwet's AC1 because kappa is

a standard nominal agreement statistic, and AC1 is less sensitive to the prevalence problem that can distort agreement estimates when one class dominates.^{47,48} The pilot produced Cohen's $\kappa = 0.617$ and Gwet's AC1 = 0.919. A third reviewer resolved disagreements during full annotation. Posts labeled *unclear* ($n = 19$) were excluded from supervised training and final metric calculation.

Quantifying community language differences

Specific examples can make the difference between communities easier to see, but examples alone cannot show whether the corpora are systematically distinct. We quantified language separation at the token level and the post level.

For the lexical analysis, we used the log-odds ratio with an informative Dirichlet prior to identify terms that distinguished the two corpora. An informative prior stabilized rare-word estimates.⁴⁹ If for token w and community c , y_{wc} denotes the token count, n_c the total token count, α_w the pooled prior count, and α_0 the total prior mass, the log-odds statistic for Type 1 relative to Type 2 was

$$\delta_w = \log\left(\frac{y_{w1} + \alpha_w}{n_1 - y_{w1} + \alpha_0 - \alpha_w}\right) - \log\left(\frac{y_{w2} + \alpha_w}{n_2 - y_{w2} + \alpha_0 - \alpha_w}\right). \quad (1)$$

The approximate variance was

$$\sigma^2(\delta_w) = \frac{1}{y_{w1} + \alpha_w} + \frac{1}{y_{w2} + \alpha_w}, \quad (2)$$

and the standardized score was $z_w = \delta_w / \sqrt{\sigma^2(\delta_w)}$. Terms with $p < 0.001$ were treated as distinctive. We ranked terms with

$$R_w = |\delta_w| \times |z_w|, \quad (3)$$

which allowed term selection to account for both effect size and estimate stability.

To evaluate post-level structure, we analyzed how complete posts separated in high-dimensional text representation space. We projected post-level embeddings into two dimensions with Pairwise Controlled Manifold Approximation and Projection, or PaCMAP.⁵⁰ We selected PaCMAP for its ability to preserve local neighborhoods and retain enough broader geometry for visual interpretation. After projection, we estimated 95% kernel density boundaries for each community and reported the Silhouette Score, Davies-Bouldin Index, and centroid distance as descriptive summaries of separation. We used these metrics only to characterize corpus structure, not as supervised classifiers or formal tests of group membership. Centroid distance was interpreted as a descriptive summary of relative displacement in the PaCMAP plane, not against a universal threshold.

Testing transfer between patient communities

A language difference has consequences for digital medicine once it changes model behaviour. To test that link, we compared two stigma detectors on the same held-out Type 1 stigma posts. One detector learned from Type 1 discourse, and the other learned from Type 2 discourse. This setup isolates

the effect of community-specific language on the model's reading of the same texts. In this study, we defined patient culture through community language use over metadata alone. We used differences in high-weight vocabulary as evidence of cultural shift, and preserved them as part of the signal being tested

We trained two L2-regularized logistic regression classifiers on Term Frequency-Inverse Document Frequency (TF-IDF) vectors. We chose a linear TF-IDF model as a transparent lexical probe instead of a higher-capacity encoder, since a more complex encoder could mix lexical transfer with hidden representation learning and make transfer failures harder to interpret. This approach made it possible to trace transfer patterns to shifts in feature weighting and term salience. We fit a single TF-IDF vocabulary on all training texts after excluding the held-out Type 1 test posts and applied it to both classifiers. This shared vocabulary kept the feature space constant across models and allowed the comparison to focus on learned feature weights instead of differences in input representation. The in-domain classifier learned from labeled Type 1 posts. The out of domain classifier learned from labeled Type 2 posts. A held-out test set of confirmed Type 1 stigma posts served as the common evaluation set, and none of these posts were used in either training set. This design compared two readings of the same Type 1 stigma texts, one learned from Type 1 discourse and one learned from Type 2 discourse.

We compared the two sets of predicted stigma probabilities with the Kolmogorov-Smirnov statistic. We also reported recall and F1 for the stigma class. Our goal was not to produce the strongest possible general classifier but to evaluate how well a model trained in one patient community could detect stigma expressed through the language of another community.

Direct LLM prompting under community calibrated prompts

The TF-IDF transfer test was designed as a transparent lexical probe. We added a direct LLM prompting test to determine if prompt context changed stigma sensitivity across Type 1 and Type 2 posts. We evaluated GPT-5.4 and Llama-3.1-8B-Instruct under four prompt settings: a generic codebook prompt, a Type 1 calibrated prompt, a Type 2 calibrated prompt, and a pooled calibrated prompt. The Type 1 calibrated prompt used only Type 1 examples and Type 1 task language. The Type 2 calibrated prompt used only Type 2 examples and Type 2 task language. The pooled prompt used examples from both communities. The same held out Type 1 and Type 2 test posts were used across prompt settings. For each model and prompt setting, we reported Macro F1, positive class F1, negative class F1, macro precision, macro recall, and accuracy. The key comparison was community matched prompting versus cross community prompting. This design evaluated how community context affected model sensitivity to stigma.

Prompt calibration for general LLMs

Prompt calibration was the first component of the culture-sensitive calibration and adjudication pipeline proposed in this study. We designed this component to improve stigma detection

without changing model weights when the stigma boundary depended on community-specific wording, ordinary treatment terms, and indirect expressions of blame or shame. We implemented prompt calibration through program-level optimization in DSPy, which represents an LLM workflow as a parameterized program and optimizes it against a downstream task metric.⁴¹

If x denotes an input post and $f_\phi(x)$ denotes the program output, then $\phi = (I, D, \pi)$ contains instruction text I , a demonstration pool D , and a retrieval policy π that selects examples for the current input when retrieval is used. For validation set $\mathcal{D}_{val}$ and metric m , prompt optimization can be written as

$$\phi^* = \operatorname{argmax}_{\phi \in \Phi} \frac{1}{|\mathcal{D}_{val}|} \sum_{(x,y) \in \mathcal{D}_{val}} m(f_\phi(x), y). \quad (4)$$

This objective matched our task, as the quantity of interest was the final label and not the quality of intermediate reasoning traces. We used GPT-5.4 as the proprietary model and Llama-3.1-8B-Instruct as the open-weight comparator.^{39,40} We recorded model identifiers, access dates, prompts, and decoding settings for reproducibility.

We compared five prompt configurations. The zero-shot baseline used the codebook definition without demonstrations. Bootstrap Few-Shot searched for demonstrations that improved performance on a validation split and reused the strongest set during inference. KNN Few-Shot retrieved the $K = 8$ most similar labeled examples for each input based on cosine similarity between the input embedding $\mathbf{e}_x$ and candidate embedding $\mathbf{e}_d$,

$$\operatorname{sim}(\mathbf{e}_x, \mathbf{e}_d) = \frac{\mathbf{e}_x \cdot \mathbf{e}_d}{\|\mathbf{e}_x\| \|\mathbf{e}_d\|}. \quad (5)$$

The retrieved demonstration set for input x was

$$D_x = \operatorname{TopK}_{d \in \mathcal{R}} \operatorname{sim}(\mathbf{e}_x, \mathbf{e}_d), \quad (6)$$

where $\mathcal{R}$ denotes the labeled retrieval pool. Bootstrap Random Search and Bootstrap Optuna searched over candidate demonstration sets. MIPROv2 optimized both instructions and demonstrations through surrogate-guided black-box search.⁴² For instruction I , demonstration set D , and validation objective S , the search target was

$$(I^*, D^*) = \operatorname{argmax}_{I, D} S(I, D \mid \mathcal{D}_{val}). \quad (7)$$

A full enumeration of candidate instructions and demonstration sets was not practical, since the number of combinations grows rapidly with prompt length and candidate examples. MIPROv2 reduces this search burden by ranking candidate programs with surrogate-guided search. It uses minibatch evaluations to screen candidates and reserves full validation passes for promising programs.⁴² This search design matched our task, where the prompt space was discrete, the program had more than one prompt component, and the metric was available only at the final task output.

For binary stigma detection, the validation objective m was Macro F1 on the full training split. For stigma subtype detection, m was multi-label Macro F1 computed only on posts labeled positive for stigma. This restriction limited the influence

Table 6: Hyperparameter configuration for DSPy optimizers.

Method	Hyperparameters
MIPROv2	num_candidates=3, init_temperature=0, max_bootstrapped_demos=8, max_labeled_demos=16, num_trials=15, minibatch_size=25, minibatch_full_eval_steps=10
Bootstrap Few-Shot	max_bootstrapped_demos=8, max_labeled_demos=16
Bootstrap Few-Shot	max_bootstrapped_demos=8, max_labeled_demos=16,
Random Search	num_candidate_programs=3
Bootstrap Few-Shot	max_bootstrapped_demos=8, max_labeled_demos=16,
Optuna	num_candidate_programs=3
KNN Few-Shot	$K = 8$

of the dominant non-stigma class and focused optimization on subtype separation. Table 6 lists the optimizer settings.

Semi-supervised adjudication and pseudo-label selection

Importance-weighted adjudication was the second component of our culture-sensitive calibration and adjudication framework. This component addressed a second technical problem in our study, as stigma posts and subtype labels were sparse, while simple pseudo-labeling can fold early model errors into the training set. We therefore expanded label coverage with an adjudication rule that treated disagreement among models as a signal of uncertainty about cultural meaning, instead of treating it as noise to average away.

The adjudication step combined three models with different training histories and model families. DepRoBERTa contributed an encoder model trained on mental health social media language. Llama 3.1 8B Instruct contributed an open-weight decoder model, and GPT-5.4 contributed a proprietary frontier LLM.^{39,40,51} For each unlabeled sample x_i , model m_j produced a probability vector q_{ij} over the binary label set $C = \{\text{no stigma}, \text{stigma}\}$. DepRoBERTa probabilities came from the softmax layer. For the LLMs, constrained label log-probabilities were converted with softmax normalization so all three outputs lay on a common class-probability scale.

The adjudication rule weighted each model at the level of the individual post and did not assign one fixed weight to the whole unlabeled pool. Let $u_c = 1/|C|$ denote the uniform label distribution. After clipping each probability away from zero with small constant ϵ and renormalizing, we measured model decisiveness for sample x_i with

$$d_{ij} = D_{\text{KL}}(q_{ij} \| u) = \sum_{c \in C} q_{ijc} \log \left(\frac{q_{ijc}}{u_c} \right). \quad (8)$$

A near-uniform prediction yields a small d_{ij} , and a concentrated prediction yields a larger value. We converted these decisiveness scores into sample-level model weights,

$$w_{ij} = \frac{d_{ij} + \epsilon}{\sum_{k=1}^J (d_{ik} + \epsilon)}, \quad (9)$$

Algorithm 1 Importance-weighted adjudication for pseudo-label selection

```

1: Input unlabeled texts  $\mathcal{U}$ , models  $m_1, \dots, m_J$ , labels  $C$ , confidence threshold  $\tau$ , margin threshold  $\Delta$ , small constant  $\epsilon$ 
2: Output accepted pseudo-labeled set  $\mathcal{P}$ 
3:  $\mathcal{P} \leftarrow \emptyset$ 
4:  $K \leftarrow |C|$ 
5: for each text  $x_i \in \mathcal{U}$  do
6:   for each model  $m_j$  do
7:     Obtain class probabilities  $q_{ijc} = P_{m_j}(y = c \mid x_i)$  for each  $c \in C$ 
8:     Clip  $q_{ijc}$  to at least  $\epsilon$  and renormalize
9:     Compute decisiveness  $d_{ij} = \sum_{c \in C} q_{ijc} \log(q_{ijc} K)$ 
10:   end for
11:   Compute weights  $w_{ij} = \frac{d_{ij} + \epsilon}{\sum_{k=1}^J (d_{ik} + \epsilon)}$ 
12:   for each class  $c \in C$  do
13:     Aggregate class score  $s_{ic} = \sum_{j=1}^J w_{ij} q_{ijc}$ 
14:   end for
15:    $\hat{y}_i \leftarrow \operatorname{argmax}_{c \in C} s_{ic}$ 
16:    $\gamma_i \leftarrow \max_{c \in C} s_{ic}$ 
17:    $\rho_i \leftarrow \gamma_i - \max_{c \in C, c \neq \hat{y}_i} s_{ic}$ 
18:   if  $\gamma_i \geq \tau$  and  $\rho_i \geq \Delta$  then
19:     Add  $(x_i, \hat{y}_i)$  to  $\mathcal{P}$ 
20:   end if
21: end for

```

and aggregated class probabilities directly,

$$s_{ic} = \sum_{j=1}^J w_{ij} q_{ijc}. \quad (10)$$

The final binary pseudo-label was

$$\hat{y}_i = \arg \max_{c \in C} s_{ic}. \quad (11)$$

We accepted a pseudo-label only when the top class probability $\gamma_i = \max_c s_{ic}$ exceeded confidence threshold τ and the gap to the second ranked class ρ_i exceeded margin threshold Δ . The confidence and margin thresholds were selected on validation data. This rule gives more influence to the model that is more decisive for a particular post. A pseudo-label still had to pass the confidence and margin thresholds before entering the expanded training set. Algorithm 1 summarizes the binary pseudo-label selection procedure.

Subtype adjudication used the same logic, but the output was multi-label and not binary. For subtype ℓ , each model produced a one-vs-rest probability $q_{ij\ell}$, and the weighted subtype score was

$$s_{i\ell} = \sum_{j=1}^J w_{ij} q_{ij\ell}. \quad (12)$$

The subtype label was assigned as

$$\hat{y}_{i\ell} = \mathbb{I}[s_{i\ell} \geq \tau_\ell], \quad (13)$$

where τ_ℓ is the validation-selected threshold for subtype ℓ . Posts assigned no stigma received an all-zero subtype vector. Figure 1 summarizes the full pipeline.

Parameter-efficient fine tuning

The fine tuning stage evaluated the effect of data selection on downstream learning under the same model and training conditions. To keep the comparison tied to the data and separate from model scale, we adapted a compact instruction model under a realistic compute budget and used the same training recipe for each dataset configuration.

We fine tuned Llama-3.2-3B on the original labeled set and on each expanded training configuration. We used parameter-efficient fine tuning with Low-Rank Adaptation (LoRA) for model adaptation under limited memory, with most pretrained weights kept fixed.^{52,53} For a pretrained weight matrix W_0 , LoRA represents the adapted matrix as

$$W = W_0 + \Delta W, \quad \Delta W = BA, \quad (14)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, and r is a low rank. This factorization reduces the number of trainable parameters and keeps updates within a restricted subspace. We set the LoRA rank to $r = 8$, the scaling parameter to $\alpha = 32$, and dropout to 0.1. The same fine tuning settings were used for the original labeled set, majority-voting pseudo-labels, ensemble-adjudicated pseudo-labels, and GPT-5.4-only pseudo-labels.

We loaded the base model in 8-bit precision to reduce memory use and trained with the AdamW8bit optimizer.⁵⁴⁻⁵⁶ This setup allowed us to fine tune Llama-3.2-3B within the available GPU budget without full precision model states. The learning rate was 5×10^{-5} , the batch size was 2, and gradient accumulation used 4 steps. The same held-out test set was used for all fine tuned models.

Evaluation and statistical analysis

Our evaluation strategy aligned with the class imbalance in the data and with the goal of detecting stigma in patient discourse. As stigma posts were uncommon, we emphasized metrics that capture minority-class performance, with accuracy reported as a secondary summary.

The detection of binary stigma was evaluated with accuracy, macro precision, macro recall, Macro F1, positive class F1, and negative class F1. Macro F1 was our primary metric, as it gives equal weight to each class and is more informative when the majority class dominates.³² The positive class was defined as any post labeled with diabetes-related stigma. The negative class was defined as no stigma. Posts labeled *unclear* were excluded from metric calculation.

Stigma subtype detection was evaluated as a multi-label task over the five subtype labels. A subtype prediction counted as correct only when the model selected the specific subtype assigned by annotators. A generic stigma prediction without the correct subtype did not count as correct for subtype evaluation. All comparisons used the same held-out test set, including prompt configurations, pseudo-labeling strategies, and fine tuned models.

We also reported type stratified performance for the final recovery framework. Diabetes type was used only after prediction to audit model performance across Type 1 and Type 2 posts. For each model setting, we computed Macro F1 and positive class

F1 separately for Type 1 and Type 2 test posts, and we reported the absolute difference between the two Macro F1 values as a simple performance gap. This analysis measured community-level detection changes after calibration and adjudicated expansion, alongside aggregate performance. For paired comparisons between models or prompt settings evaluated on the same held out examples, we used the exact McNemar test on paired correctness. Each example was coded as correct or incorrect for each model, producing a paired 2×2 table of discordant predictions. For binary stigma detection, correctness was defined by the binary stigma label. For multi-label subtype detection, correctness was defined as exact agreement with the annotated subtype vector, and we treated these tests as secondary, as subtype labels were sparse and overlapping.

Patient narrative check

As the manually annotated reference set came from condition specific online forums, we added a small patient narrative check using the HealthUnlocked forums for diabetic communities.⁵⁷ We selected 150 published patient narratives for manual review, of which 75 belong to the Type 1 diabetes category and the others to the Type 2 diabetes category. Two annotators applied the same stigma codebook used for the forum corpus. These labels were used only for the patient narrative check and were not used to tune prompts, select pseudo labels, or train the final model. This analysis measured how well the calibrated model pipeline maintained signal in patient language outside the forum reference set.

Ethics and privacy

This secondary analysis did not require Research Ethics Board review under institutional guidance. The data consisted of public online forum posts, published patient narratives, and interview excerpts that were deidentified after receipt and before analysis. No new participants were recruited or contacted for this analysis, and no identifiable private information was analyzed. Usernames, profile links, URLs, and direct identifiers were removed from analytic outputs. Raw interview excerpts are not publicly released. Results are reported only in aggregate and direct quotations were avoided unless needed to explain a linguistic pattern.

Data Availability

The annotation codebook, labels, pseudo labels, prompts, model outputs, evaluation splits, and scripts needed to reproduce the reported analyses are available at <https://github.com/HIVE-UofT/Diabetes-Stigma-Detection-Framework>. For public forum data, we provide post identifiers and reconstruction scripts where permitted by platform terms and user privacy considerations. Raw deidentified interview excerpts are not publicly released because of participant privacy restrictions. Published patient narratives are referenced through their public sources where redistribution is permitted. All restrictions are described in the repository.

Code Availability

Implementations proposed in this study, such as prompt optimizations in the DSPy framework, the semi-supervised learning on unlabeled samples, and fine tuning of LLMs, are available at <https://github.com/HIVE-UofT/Diabetes-Stigma-Detection-Framework>. The weights for the fine tuned models are available at <https://huggingface.co/haji80mr-uoft>.

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Author contributions

A.H.M.R. implemented the machine learning models and proposed algorithms, evaluated and interpreted the results, and contributed to the writing. M.S. contributed to the writing and qualitative analysis of the results. S.A. and D.F. contributed to the conceptualization of the stigma constructs, discussion of findings, and preparation of the study data. K.S. and M.N. provided feedback across different phases of model implementation and qualitative analysis. M.N., D.F., and N.S. critically reviewed the manuscript and provided feedback. Z.S. conceptualized and designed the study, supervised the project, contributed to the model analysis, implementation, visualizations, and writing, and critically reviewed the manuscript.

Competing Interests

The authors declare no competing interests. Also, the funders of the study had no role in study design, data collection and analysis, or interpretation of results and preparation of the manuscript.

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